
LinkedIn — [linkedin.com/in/ritvik-ravi](https://www.linkedin.com/in/ritvik-ravi)

Education

Bachelor of Technology — Mechanical Engineering
Indian Institute of Technology Tirupati, India

Expected: May 2027

Skills

CFD & Structural Simulation: ANSYS Fluent, ANSYS APDL (FEA/FEM), XFOIL

CAD & Design: SolidWorks, Fusion 360, AutoCAD

Programming: Python (NumPy, Pandas, Scikit-learn, OpenCV), MATLAB, C, ROS2

ML & Data: Regression, Classification, Clustering, Feature Engineering, Pattern Recognition, CNNs

Domains: Aerodynamics, Fluid Mechanics, Control Systems, Structural Analysis, Propulsion Systems, UAV Design, Design Optimization

Projects

LAT Aerospace Wing Optimization — Inter IIT Tech Meet (High Prep) Oct 2025 – Dec 2025
Team Lead

- Spearheaded a multidisciplinary team solving a high-prep aerospace problem statement focused on a **propulsion-augmented high-lift wing system**.
- Engineered CFD-based aerodynamic models to analyze aero-propulsion integration, achieving a **Lift-to-Drag (L/D) ratio of 11** and a **Lift Coefficient (CL) of 7.2**.
- Optimized propulsion distribution using 4 EDF units to **generate 92.6N total thrust, exceeding the 37N mission requirement** for enhanced climb performance.
- Executed structural validation in ANSYS APDL, performing mesh convergence analysis (<5% error) to confirm a **safety factor of 161 under maximum aerodynamic loads (V_D)**.
- Formulated a business case and unit economic model, **reducing CASK** (Cost per Available Seat Kilometer) to **26.1 INR, a 42% improvement** over legacy turboprop benchmarks.

Small-Range Rocket Design and Propulsion Optimization

Nov 2023 – May 2024

Group Project

- Engineered a **Class-K solid rocket motor** (1449 Ns impulse) with a **200mm propellant grain**, utilizing **OpenMotor** and isentropic nozzle theory for internal ballistics analysis.
- Achieved a **delivered thrust coefficient (C_f) of 1.51** and **80.16% volumetric loading**, optimizing nozzle expansion to sustain a **1.80s burn time** for stable flight profiles.
- Calculated and refined grain geometry to maximize the propellant-to-case ratio, resulting in a design recognized for excellence by the Institute's Director and showcased at Tirutsava, the institute's annual fest.

Star Detection and Constellation Identification App

Jan 2024 – May 2024

Group Project

- Engineered an **end-to-end AI/ML pipeline** to automate stellar centroid detection from high-noise, user-captured astronomical images.
- Implemented a custom pattern-matching algorithm using **distance-vector matching and nearest-neighbor search (k-NN)** to identify constellations against a dataset of 1200+ stellar images with 88% classification accuracy.
- Optimized classification logic for real-time identification, streamlining the data flow from raw pixel input to celestial coordinate mapping.

Technical Portfolio & Systems Analysis (ritvikravi.space)

Jan 2026 – Present

Independent Researcher & Writer

- **Authoring a technical repository** of research-driven case studies and engineering essays, focusing on **first-principles analysis** of complex system trade-offs.
- **Developing cross-disciplinary frameworks**, such as a "Mechanical Engineering Hedge Fund" model and "Project Design in the AI Era," to evaluate the **intersection of technical feasibility and business strategy**.
- **Synthesizing complex data** across aerodynamics, manufacturing, and failure analysis to produce **consulting-style whitepapers** on technology trends and historical engineering shifts.

Experience

AlmaNest — CEO & Founder

Aug 2024 – Present

- Founded a student-focused digital platform **incubated under IIT Tirupati's Institution Innovation Council (IIC)**.
- Directed the Product Lifecycle (PLC) from initial design to market entry; scaled operations by leading a multidisciplinary team, driving the execution strategy for a high-traffic environment under a technical incubation framework.
- Secured **INR 1.5 lakh** in seed funding; piloted a festival-scale deployment with 600+ users.

Entrepreneurship Cell (E-Cell), IIT Tirupati

Event Coordinator

Sep 2024 – May 2025

Deputy Head

Aug 2025 – Present

- Coordinated event logistics and execution for institute-wide hackathons and E-Summit 2026, which drew over **450 participants** nationwide, resulting in a 100% increase in year-over-year engagement metrics.
- Directed cross-functional teams across sponsorships, operations, and PR; optimized internal workflows to enhance team productivity and event delivery.

TEDx IIT Tirupati

Production Lead

Jan 2025 – Feb 2025

Co-Organizer (Lead Organizer)

Dec 2025 – Present

- Orchestrated complex production logistics and stage design for TEDx 2025, managing on-ground operations and technical workflows for a fixed audience of **240 attendees**.
- As co-organizer, spearheaded end-to-end strategic planning and execution for the 2026 edition, overseeing speaker curation, team coordination, and stakeholder management to ensure brand alignment and meeting strict TED licensing requirements.
- Facilitated high-level speaker relations, managing the onboarding process and speech development for a diverse roster of industry leaders and innovators.

Awards

1. 1st Place — ISRO-SHAR Statewide Space Quiz (World Space Week), 2020
2. 3rd Place — Stratethon Business Case Competition, IIT Dhanbad
3. 1st Place — Space Quiz, IIT Tirupati & IISER Tirupati (AEthereum Fest)

Participation

- NSSC 2023 — Team Lead (Data Analytics Competition); finalist among **15 national teams**
- Kotler's Maniac, IIT Varanasi (Case Study Competition)— Solo participant; Top **8 of 386**
- Dreams to Dollars, GLIM Chennai (Case Study Competition)— Solo finalist; **3rd place among top 4 teams**